

P3: no  $Q = -1$  relation for  $U$  and  $V$

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Labels: PROVED means a complete argument, given the inputs it cites. VERIFIED means high-precision floating-point numerics, not interval arithmetic. HEURISTIC means an argument without error control. Notation follows `paper/journal/paper2a.tex` (“paper2a”) and `rootunity_zeros/general/P1f/P1` (“P1f”).

## Result

**Theorem 1** (PROVED, with the standing of Corollary `cor:arcUV`). *Let  $f = U$  or  $f = V$ . There are no polynomials  $A_0, A_1, b \in \mathbb{C}[x]$  with  $(A_0, A_1) \neq (0, 0)$  and*

$$A_0(x)f(x) + A_1(x)f(-x) = b(x).$$

*Hence no relation  $\sum_i a_i(x)f((-1)^i x) = b(x)$  holds unless  $\sum_{i \text{ even}} a_i = \sum_{i \text{ odd}} a_i = 0$  (paper2a, Remark `rem:rootunity`, reduction).*

*Standing.* The proof uses the poles of  $f$  near  $\pm\sqrt{\zeta}$  for infinitely many roots of unity  $\zeta = e(a/N)$  with  $N$  odd, together with the limits of  $t_1$  and of the pairings there. These come from paper2a Corollary `cor:arcUV`, i.e. P1f Theorem C and the main theorem of `P1g_lemma.tex`. So the standing is that of Theorems `thm:model` and `thm:RJ`, plus the computer-assisted inputs of those results (the Arb certificates of P1e and P1g and the landscape bounds of P1f). Corollary `cor:arcUV` now covers  $\frac{a}{N} \in [\frac{1}{6}, \frac{5}{6}]$  (after P1h); the proof only needs some infinite family of odd-order accumulation points, and  $\frac{a}{N} \in [\frac{1}{5}, \frac{4}{5}]$ , for example  $a = \frac{N-1}{2}$ , already suffices, so this update does not weaken anything. Nothing else is computer-assisted: the new steps (Lemmas 5–8 and §4) are proved by hand.

*Idea.* Comparing pole sets gives nothing at  $Q = -1$  (§1). What a relation does force is an *arithmetic* constraint. At a root of unity  $\zeta$  of odd order  $N$ , the limit  $t_1^*(\zeta)$  must lie in a quadratic extension of  $K = \mathbb{Q}(\zeta)$  of the form  $K(\sqrt{\kappa})$ , where  $\kappa$  is the value at  $\sqrt{\zeta} \in K$  of a fixed rational function. We show that  $t_1^*(\zeta)$  generates  $K(\sqrt{D_N})$  with  $D_N = c^N(c^N + 4)$ ,  $c = -2(1 - \zeta)$ . At the primes above 2,  $D_N$  has odd valuation and  $\kappa$  has even valuation. The quadratic irrationality is carried by the root  $w$  of  $c^N w = (1 - w)^2$ . In the complex embedding  $w$  is exponentially small,  $|w| \leq 2|c|^{-N}$ . This is the “exponentially small information” of the programme row, read 2-adically.

## 1 Why the pole sets and the two limit points alone do not decide

**Proposition 2** (PROVED, same standing). *For  $Q = -1$  the successor argument of paper2a Proposition `prop:noqdiff` gives nothing. Every pole  $p$  of  $f$  with  $q^* = p^2$  a zero of  $1 - \Sigma_1$ ,  $S_e(q^*) \neq 0$ ,  $1 - \mathfrak{g}t_1 \neq 0$  and  $\Pi(\pm p) \neq 0$  has  $-p$  as a pole of the same order. The families near  $\sqrt{\zeta}$  and  $-\sqrt{\zeta}$  are the two square roots of one family  $q_j^*$ . They have the same  $t_j^*$ , the same tie ray, the same spacing  $2\pi/|\Delta V|$  and the same  $V$ -values.*

*Proof.* The resolvent,  $\Psi$ ,  $t_1$  and  $B_0$  depend on  $x$  only through  $q = x^2$  (proof of `prop:minusx`). The local form of `prop:shape` at a non-real zero (proof of `cor:Uminusone`) gives order  $\pi$  at both roots once  $\Pi(\pm p) \neq 0$ .  $\square$

So the only information is in the leading Laurent coefficients. At  $x = \pm i$  ( $q \rightarrow -1$ ) they give nothing either: by `cor:Uminusone`,  $\Pi_q \rightarrow -4$  at *both* roots, whatever the value of  $t_1 \asymp 1$ . The ratio  $\rho$  below tends to 1 and its first-order behaviour needs the unproved  $O(t)$  term of  $t_1$  (HEURISTIC  $t_1 = \phi - \frac{\phi}{10}t$ ). At  $q = i$  the limit  $t_1^* = \frac{(\sqrt{15}-3)-(\sqrt{15}+3)i}{12}$  lies in  $\mathbb{Q}(\zeta_8)(\sqrt{15})$ , a quadratic extension of  $\mathbb{Q}(\sqrt{i})$ . This is exactly what a relation allows (Lemma 4), so one point cannot refute it. The argument therefore needs infinitely many roots of unity of *odd* order, where  $\sqrt{\zeta} \in \mathbb{Q}(\zeta)$  and 2 is unramified. The arc supplies them.

## 2 The residue identity at a non-real pole

**Lemma 3** (PROVED). *If a relation as in Theorem 1 exists, one exists with  $A_0, A_1, b \in \mathbb{Z}[x]$  and  $(A_0, A_1) \neq (0, 0)$ .*

*Proof.*  $f(x), f(-x) \in \mathbb{Z}[[x]]$ . For a degree bound  $d$ , the relation is a homogeneous linear system with integer coefficients in the coefficients of  $(A_0, A_1, b)$ . A non-zero complex solution gives a non-zero rational one. Scale it to be integral. If  $A_0 = A_1 = 0$  then  $b = 0$ , so  $(A_0, A_1) \neq 0$ .  $\square$

**Lemma 4** (PROVED, same standing). *Assume a relation with  $A_i \in \mathbb{Z}[x]$ . Let  $\zeta = e(a/N)$  be in the family of `cor:arcUV`, let  $x_0$  be either square root of  $\zeta$ , and let  $t_1^* = t_1^*(\zeta)$  be the limit of P1f Theorem C(ii). Put*

$$\beta_{\pm} = t_1^* \frac{1 \mp x_0 + \zeta}{1 + \zeta} \pm \frac{1}{2x_0}, \quad \gamma_+ = t_1^* + \frac{1 + x_0}{2x_0}, \quad \gamma_- = t_1^* - \frac{1 - x_0}{2x_0}.$$

*Then*

$$\text{for } f = U : A_0(x_0)(1+x_0)^4\beta_+^2 = A_1(x_0)(1-x_0)^4\beta_-^2, \quad \text{for } f = V : A_0(x_0)(1+x_0)^2\gamma_+^2 = A_1(x_0)(1-x_0)^2\gamma_-^2,$$

*and  $\beta_{\pm}, \gamma_{\pm} \neq 0$ . Moreover  $A_0 \not\equiv 0$ ,  $A_1 \not\equiv 0$  and  $A_0(x)A_0(-x) = A_1(x)A_1(-x)$ .*

*Proof.* Let  $q_j^* \rightarrow \zeta$  be the zeros of Theorem `thm:arc`, and let  $p = p_j \rightarrow x_0$  be the root of  $x^2 = q_j^*$  near  $x_0$ . By `prop:shape` at a non-real zero (proof of `cor:Uminusone`),  $f = \mathcal{N}(q - q_j^*)^{-\pi} + \dots$  with  $\mathcal{N} = c_*\Pi(x)^2/(2x)$ . The factors  $c_* \neq 0$  and  $\pi \geq 1$  depend on  $q_j^*$  only. Since  $q - q_j^* = (x - p)(x + p)$ , the leading coefficients are  $N_+ = c_*\Pi(p)^2/(2p) \cdot (2p)^{-\pi}$  at  $x = p$ , and  $c_*\Pi(-p)^2/(-2p) \cdot (-2p)^{-\pi}$  at  $x = -p$ . The function  $f(-x)$  has leading coefficient  $N_- = c_*\Pi(-p)^2/(-2p) \cdot (2p)^{-\pi}$  at  $x = p$ . The polynomials  $A_i$  and  $b$  are holomorphic, so the coefficient of  $(x - p)^{-\pi}$  in the relation gives  $A_0(p)\Pi(p)^2 = A_1(p)\Pi(-p)^2$ . The same step at  $-p$  gives  $A_0(-p)\Pi(-p)^2 = A_1(-p)\Pi(p)^2$ . Here  $\Pi = \Pi_q$  for  $U$  and  $\Pi = \Pi_1$  for  $V$ . For large  $j$  the pairings at  $\pm p$  are non-zero (`cor:arcUV`). If  $A_1 \equiv 0$ , then  $A_0(p_j) = 0$  for all large  $j$ , so  $A_0 \equiv 0$ ; the same holds with the roles exchanged. Multiplying the two identities gives  $A_0A_0(-\cdot) = A_1A_1(-\cdot)$  at the infinitely many distinct  $p_j$ , hence identically. Now insert the closed forms `eq:RJclosedq` and `eq:RJclosed1`, which hold at  $q_j^*$  for both roots (proof of `cor:Uminusone`). For  $U$ ,  $\Pi_q(\pm p) = \frac{2qB_0(1\pm p)}{1\mp p} (t_1^{\frac{1\mp p+q}{1+q}} \pm \frac{1}{2p})$ . The factor  $2qB_0$  is common to both roots and cancels. For  $V$ ,  $(1 - \mathbf{g}_V t_1)\Pi_1(\pm p) = \frac{2q(1\pm p)}{1-q} (t_1 \pm \frac{1\pm p}{2p})$ , and  $(1 - \mathbf{g}_V t_1)$  and  $2q/(1 - q)$  cancel. Let  $j \rightarrow \infty$ . Then  $t_1(q_j^*) \rightarrow t_1^*$  (P1f Thm. C(ii)),  $B_0$  and  $(1 - \mathbf{g}_V t_1)^{-1}$  have finite non-zero limits, and the brackets tend to  $\beta_{\pm}, \gamma_{\pm}$ , which are non-zero by the  $\omega$ -conditions of P1f Thm. C(iii) (certified in P1g). This gives the displayed identities.  $\square$

### 3 Arithmetic of $t_1^*$ at odd order

Fix  $N \geq 5$  odd,  $\gcd(a, N) = 1$ ,  $\zeta = e(a/N)$ ,  $K = \mathbb{Q}(\zeta)$ . As in P1f §2,  $c = -2(1 - \zeta)$ ,  $X = c^N$ ,  $w$  is the root of  $w^2 - (2 + X)w + 1 = 0$  (that is,  $c^N w = (1 - w)^2$ ) with  $|w| < 1$ ,  $\lambda = (1 - w)^{1/N}$  (principal),  $u_0 = \lambda^2/c$ , and  $q := \zeta^{-1}$  inside  $q$ -Pochhammer symbols in this section only. P1f Theorem C(ii) gives  $t_1^* = -\frac{1}{2}(1 + \omega)$  with  $\omega = Z_N^-(u_0 Z_N^+)$ ,  $Z_N^\pm = \sum_{k \geq 0} \psi_k \zeta^{-k^2 \mp k}$ ,  $Z_N^+ \neq 0$ .

**Lemma 5** (finite form; PROVED). *For  $y \in \mathbb{C}$  with  $y^N = 1 - w$  put  $u(y) = y^2/c$  and*

$$T^\pm(y) = \sum_{s=0}^{N-1} \frac{\zeta^{-s^2 \mp s} u(y)^s}{(yq; q)_s}, \quad \Omega(y) = \frac{T^-(y)}{u(y) T^+(y)}.$$

*Then  $Z_N^\pm = f_0(w) T^\pm(\lambda)$  with  $f_0(w) \neq 0$ , and  $\omega = \Omega(\lambda)$ .*

*Proof.*  $e^{\Phi(Y)} = F(u_0 Y)$  with  $F = \exp H$ ,  $H(v) = \sum_{n \nmid N} v^n / (n(1 - \zeta^{-n}))$ . So  $\psi_k = h_k u_0^k$ , where  $F = \sum h_k v^k$  converges on  $|v| < 1$  and  $|u_0| < 1$ . Write  $F(v) = \sum_{s < N} v^s g_s(v^N)$ . The functional equation  $(1 - \zeta v)F(\zeta v) = (1 - v^N)^{1/N} F(v)$  (S0 Lemma tel, PROVED), compared in the class  $v^{s \bmod N}$ ,  $1 \leq s \leq N-1$ , gives  $g_s = g_{s-1} / (1 - \lambda(W) \zeta^{-s})$ ,  $W = v^N$ . Hence  $g_s = g_0 / (\lambda(W) q; q)_s$ . Since  $\zeta^{-k^2 \mp k}$  depends only on  $k \bmod N$ , and the sums converge absolutely,  $Z_N^\pm = \sum_{s < N} \zeta^{-s^2 \mp s} u_0^s g_s(u_0^N)$ . Here  $u_0^N = (1 - w)^2 / c^N = w$  and  $\lambda(w) = \lambda$ . So  $Z_N^\pm = g_0(w) T^\pm(\lambda)$ , and  $g_0(w) \neq 0$  because  $Z_N^+ \neq 0$ . VERIFIED:  $\Omega(\lambda)$  agrees with the series value of  $\omega$  to working precision at  $\frac{1}{4}, \frac{1}{3}, \frac{2}{5}$  (P3\_conj.py, P3\_explore.py).  $\square$

**Lemma 6** (Galois invariance; PROVED).  $\Omega(\zeta^j y) = \Omega(y)$  for every  $j$  and every  $y$  with  $y^N = 1 - w$ . Consequently  $\omega \in K(w)$ .

*Proof.* Extend  $(yq; q)_n$  to  $n \in \mathbb{Z}$  by  $(yq; q)_n = (yq; q)_{n-1} (1 - y\zeta^{-n})$ . All factors are non-zero because  $y^N \neq 1$ . Then  $(yq; q)_{n+N} = (yq; q)_n (1 - y^N)$ . The summand  $\tau_s = \zeta^{-s^2 \mp s} u^s / (yq; q)_s$  is  $N$ -periodic because  $u^N / (1 - y^N) = (1 - w)^2 / (c^N w) = 1$ . Next,  $u(\zeta^j y) = \zeta^{2j} u(y)$  and  $(\zeta^j y q; q)_s = (yq; q)_{s-j} / (yq; q)_{-j}$ . With  $s' = s - j$  we have  $-s^2 \mp s + 2js = -s'^2 \mp s' + j^2 \mp j$ , hence  $T^\pm(\zeta^j y) = \zeta^{j^2 \mp j} u^j (yq; q)_{-j} T^\pm(y)$ . So  $T^-/T^+$  is multiplied by  $\zeta^{2j}$ , and so is  $u$ .  $\Omega$  is therefore invariant. The field  $K(w, \lambda)$  is a Kummer extension of  $K(w)$ , since  $\zeta \in K$  and  $\lambda^N = 1 - w$ . Its automorphisms are  $\lambda \mapsto \zeta^j \lambda$ .  $\Omega$  has coefficients in  $K$ , so  $\omega = \Omega(\lambda)$  is fixed by them and lies in  $K(w)$ . VERIFIED:  $\max_j |\Omega(\zeta^j \lambda) - \Omega(\lambda)| < 10^{-398}$  at 400 digits ( $N = 3, \dots, 11$ , P3\_check.py).  $\square$

Fix a prime  $\mathfrak{p} \mid 2$  of  $K$ , and let  $v$  be the valuation with  $v(2) = 1$  on an algebraic closure of  $K_{\mathfrak{p}}$ . Since  $N$  is odd, 2 is unramified in  $K$  and  $1 - \zeta^i$  ( $N \nmid i$ ) is a 2-unit. So  $v(c) = 1$ .

**Lemma 7** (PROVED).  $D_N := X(X + 4)$  has  $v_{\mathfrak{p}}(D_N) = N + 2$ , which is odd. Hence  $w \notin K$ ,  $K(w) = K(\sqrt{D_N})$  is quadratic, and the non-trivial  $\sigma \in \text{Gal}(K(w)/K)$  sends  $w \mapsto 1/w$ .

*Proof.*  $v(X) = N$ , and  $v(X + 4) = 2$  because  $N \geq 3$ . The discriminant of  $w^2 - (2 + X)w + 1$  is  $D_N$ .  $\square$

**Lemma 8** (separation; PROVED).  $v(\omega - \sigma\omega) = \frac{3}{2}$ . In particular  $\omega \notin K$ , and  $t_1^* \in K(w) \setminus K$ .

*Proof.* Embed  $K(w, \lambda)$  in  $\overline{K_{\mathfrak{p}}}$ . First,  $v(w) = 0$ : from  $w^2 - (2 + X)w + 1 = 0$  we get  $w + 1/w = 2 + X$ , and  $v(2 + X) = 1$  since  $v(X) = N \geq 5 > 1 = v(2)$ . If  $v(w) \neq 0$  then  $w$  and  $1/w$  have valuations of opposite sign (as  $v(w) + v(1/w) = 0$ ), so  $v(w + 1/w) = \min(v(w), v(1/w)) < 0 \neq 1$ , a contradiction; hence  $v(w) = 0$  exactly. Now write  $w = 1 - \delta$ . From  $(w - 1)^2 = Xw$  we get  $v(\delta^2) = v(X) + v(w) = N$ , so  $v(\delta) = N/2$ . The principal 2-adic root  $w^{1/N} = \sum_k \binom{1/N}{k} (-\delta)^k$  converges ( $N$  is a 2-adic unit)

and satisfies  $w^{1/N} \equiv 1$  modulo valuation  $N/2$ . Take  $u_0 := w^{1/N}$  and let  $\lambda$  be the square root of  $cu_0$  with  $\lambda^N = 1 - w$ : we have  $(cu_0)^N = c^N w = (1 - w)^2$ , and since  $N$  is odd one of  $\pm\sqrt{cu_0}$  works. By Lemma 6,  $\omega = \Omega(\lambda)$  for this root. Here  $v(\lambda) = \frac{1}{2}$  and  $u(\lambda) = u_0$ . Apply  $\sigma$ , extended to  $K(w, \lambda)$ . It sends  $\lambda$  to a root  $\lambda'$  of  $Y^N = 1 - 1/w = -\delta/(1 - \delta)$ , and by Lemma 6 we may take  $\lambda' = -\lambda(1 - \delta)^{-1/N} = -\lambda(1 + \varepsilon)$  with  $v(\varepsilon) \geq N/2$ .

With  $u_0$  held fixed, put  $\hat{T}^\pm(y) = \sum_s \zeta^{-s^2 \mp s} u_0^s / (yq; q)_s$ . For  $v(y) > 0$  each factor  $(1 - y\zeta^{-i})^{-1}$  is a geometric series with integral coefficients, so  $\hat{T}^\pm(y) = P^\pm(y^2) + yQ^\pm(y^2)$  with  $P^\pm, Q^\pm$  integral. Also  $P^\pm(0) = A^\pm := \sum_s \zeta^{-s^2 \mp s} u_0^s$  and  $Q^\pm(0) = B^\pm := \sum_s \zeta^{-s^2 \mp s} u_0^s e_s$ , where  $e_s = \sum_{i=1}^s \zeta^{-i}$ . Exactly,

$$\hat{T}^-(y)\hat{T}^+(-y) - \hat{T}^-(-y)\hat{T}^+(y) = 2y(Q^-P^+ - P^-Q^+)(y^2).$$

So  $\Omega(\lambda) - \Omega(-\lambda)|_{u_0 \text{ fixed}} = 2\lambda(Q^-P^+ - P^-Q^+)(\lambda^2)/(u_0\hat{T}^+(\lambda)\hat{T}^+(-\lambda))$ . Replacing  $-\lambda, u_0$  by  $\lambda', u(\lambda') = u_0(1 + \varepsilon)^2$  changes  $\Omega$  by valuation at least  $N/2$ , since every quantity is integral and the denominators are units (below).

*Gauss sums.* Let  $h = 2^{-1} \bmod N$  and  $G_0 = \sum_{s \bmod N} \zeta^{-s^2}$ . Completing the square,  $\sum_s \zeta^{-s^2 + bs} = \zeta^{b^2 h^2} G_0$ . So  $G^\pm := \sum_s \zeta^{-s^2 \mp s} = \zeta^{h^2} G_0$ , and these sums are equal. Using  $e_s = \zeta^{-1}(1 - \zeta^{-s})/(1 - \zeta^{-1})$ , which is  $N$ -periodic,

$$(1 - \zeta^{-1}) \sum_s \zeta^{-s^2} (\zeta^s - \zeta^{-s}) e_s = \zeta^{-1} [\zeta^{h^2} G_0 - G_0 - \zeta^{h^2} G_0 + \zeta G_0] = (1 - \zeta^{-1}) G_0,$$

so  $B_0^- - B_0^+ = G_0$ , where  $B_0^\pm = \sum_s \zeta^{-s^2 \mp s} e_s$ . Since  $G_0 \bar{G}_0 = N$  is odd,  $G_0$  is a 2-unit.

*Valuations.* Since  $u_0 \equiv 1$ ,  $A^\pm \equiv G^\pm$  and  $B^\pm \equiv B_0^\pm$  modulo valuation  $N/2$ . So  $\hat{T}^\pm(\pm\lambda) \equiv \zeta^{h^2} G_0$  modulo valuation  $\frac{1}{2}$ , which is a unit, and

$$(Q^-P^+ - P^-Q^+)(\lambda^2) \equiv B_0^- G^+ - G^- B_0^+ = \zeta^{h^2} G_0^2 \pmod{v \geq 1},$$

which is also a unit. Hence  $v(\Omega(\lambda) - \Omega(-\lambda)) = 1 + \frac{1}{2} = \frac{3}{2}$ , while the corrections have valuation at least  $\min(\frac{5}{2}, \frac{N}{2}) > \frac{3}{2}$  for  $N \geq 5$ . So  $v(\omega - \sigma\omega) = \frac{3}{2}$ . Finally  $t_1^* = -\frac{1}{2}(1 + \omega)$ .

VERIFIED (P3-check.py, 400 digits):  $G^+ = G^-$  and  $B_0^- - B_0^+ = G_0$  to  $10^{-399}$ . The norm  $N_{K(w)/\mathbb{Q}}(\omega - \sigma\omega)$  is identified as a rational number (fit error  $< 10^{-393}$ ), and its 2-adic valuation equals  $3\varphi(N)$ , which is what  $v = \frac{3}{2}$  at every  $\mathfrak{p} \mid 2$  predicts ( $e = 2$ ): 6, 12, 18, 18, 30 for  $N = 3, 5, 7, 9, 11$ . At  $N = 3$  this holds as well, although the proof needs  $N \geq 5$ .  $\square$

## 4 Proof of Theorem 1

Suppose a relation exists. By Lemmas 3 and 4,  $A_0, A_1 \in \mathbb{Z}[x]$  are both non-zero and  $A_0(x)A_0(-x) = A_1(x)A_1(-x)$ . Write  $A_i = 2^{e_i} \tilde{A}_i$  with  $\tilde{A}_i \in \mathbb{Z}[x]$  and content prime to 2. By Gauss's lemma, the 2-part of the content of  $A_i(x)A_i(-x)$  is  $2^{2e_i}$ . Hence  $e_0 = e_1$ . The reductions  $\bar{A}_i \in \mathbb{F}_2[x]$  of  $\tilde{A}_i$  are non-zero and have finitely many roots in  $\mathbb{F}_2^\times$ , each a root of unity of some odd order. Let  $M$  be the largest of these orders.

Choose  $N$  odd,  $N \geq \max(17, M + 1)$ , and  $a = \frac{N-1}{2}$ . Then  $\gcd(a, N) = 1$  and  $\frac{a}{N} \in [\frac{1}{5}, \frac{4}{5}]$ , so  $\zeta = e(a/N)$  is in the family of `cor:arcUV`, and Lemma 4 applies. Take  $x_0 = \zeta^{(N+1)/2} \in K$ , a square root of  $\zeta$ . Let  $\mathfrak{p} \mid 2$ . The reduction of  $x_0$  modulo  $\mathfrak{p}$  has exact order  $N > M$ , so  $\bar{A}_i(\bar{x}_0) \neq 0$ , and  $v_{\mathfrak{p}}(A_i(x_0)) = e_i$ . The elements  $1 \pm x_0$  are 2-units, because  $x_0$  and  $-x_0$  are roots of unity of order  $N$  and  $2N$ , and  $1 + x_0 = (1 - x_0^2)/(1 - x_0)$ . So

$$\kappa_U := \frac{A_0(x_0)(1 + x_0)^4}{A_1(x_0)(1 - x_0)^4}, \quad \kappa_V := \frac{A_0(x_0)(1 + x_0)^2}{A_1(x_0)(1 - x_0)^2}$$

are elements of  $K^\times$  with  $v_p = e_0 - e_1 = 0$ .

For  $U$ , Lemma 4 gives  $m^2 = \kappa_U$  with  $m = \beta_-/\beta_+ = M(t_1^*)$ . Here  $M$  is the Möbius map  $t \mapsto (t \frac{1+x_0+\zeta}{1+\zeta} - \frac{1}{2x_0}) / (t \frac{1-x_0+\zeta}{1+\zeta} + \frac{1}{2x_0})$ , with coefficients in  $K$  and determinant  $1/x_0 \neq 0$ . By Lemma 8,  $t_1^* \in K(w) \setminus K$ , so  $m \in K(w) \setminus K$ . Therefore  $K(\sqrt{\kappa_U}) = K(m) = K(w) = K(\sqrt{D_N})$ , and  $\kappa_U D_N \in K^{\times 2}$ . But  $v_p(\kappa_U D_N) = 0 + N + 2$  is odd (Lemma 7), a contradiction. For  $V$  the same argument applies with  $m = \gamma_-/\gamma_+$ , the Möbius map  $t \mapsto (t - \frac{1-x_0}{2x_0}) / (t + \frac{1+x_0}{2x_0})$  of determinant  $1/x_0$ , and  $\kappa_V$ .  $\square$

*Remark 9.* Only one suitable  $N$  is used, but it has to exceed a bound  $M$  that depends on the unknown  $A_i$ . That is why an infinite supply of odd-order accumulation points is needed. The  $q = -1$  and  $q = i$  families (`cor:Uminusone`, `cor:Ui`) are not used, for the reasons in §1. The real travel poles are not used either. The proof does not need the degree-9 numerics of `rem:rootunity`. It needs no bound on  $\deg A_i$  and no information on  $\rho_m$  at the real poles.

## 5 Scope, and the weakest points

- **Standing.** The proof inherits everything `cor:arcUV` rests on: Theorems `thm:model`, `thm:RJ`, the non-real use of `prop:shape` and of the closed forms (a re-reading of proofs, checked numerically in paper2a), P1f Theorem C (the formula for  $\omega$ , including the Gauss-sum ratio  $G^*(\mu + a)/G^*(\mu - a) = e(-\mu/N)$ ), and the Arb certificates of P1g ( $Z_N^+ \neq 0$  and the six  $\omega$ -margins on  $[\frac{1}{5}, \frac{4}{5}]$ ). For  $G_0^T$  nothing new is claimed. Part (iii) of `prop:noqdiff` already settles  $Q = -1$  in  $q$ .
- **New hand proofs.** Lemmas 5–8 and §4 are proved by hand only. They have not been reviewed adversarially. The 2-adic step (Lemma 8) is the most delicate: it relies on choosing the branch  $u_0 = w^{1/N}$ , which Lemma 6 permits, and on the integrality bookkeeping. It is cross-checked by the norm valuations above. That check is VERIFIED (floating point with rational identification), not certified.
- **Branch conventions.** Lemma 5 uses the principal  $\lambda$  of P1f. Lemma 6 makes  $\omega$  independent of the branch, so no convention enters the arithmetic.
- **Not covered.** Other roots of unity  $Q$  of order  $\geq 3$  are not treated. Their reduction in `rem:rootunity` needs control of  $g$ ,  $S_e$  and  $1 - \mathbf{gt}_1$  at  $\zeta^{2j} q_m$ , which is still missing. Relations with non-polynomial (for example holomorphic) coefficients are not covered, and neither is  $D$ -algebraicity.

**Files** (in `rootunity_zeros/general/P3/`; all light, mpmath floating point, each under `perl -e 'alarm 290; exit 1'` under 100 MB). `P3_explore.py` ( $\omega$  along  $a/N \rightarrow \text{seed}$ ; it shows that  $\omega$  varies continuously, so a discontinuity route fails), `P3_degree.py` (minimal polynomials of  $\text{Re } t_1^*$ ), `P3_field.py` (PSLQ:  $t_1^* \in \mathbb{Q}(\zeta, w)$  at  $\frac{1}{3}, \frac{2}{5}, \frac{3}{8}$ ), `P3_conj.py` (the finite form,  $\omega$  against  $\sigma\omega$ , branch independence), and `P3_check.py` (Gauss identities, invariance, norm valuations). No heavy computation was needed, so no Rust or Arb run was made. The only computer-assisted inputs are the inherited ones. `P3_field.py` and `P3_degree.py` take the truncation order  $K$  as an explicit argument; since  $|u_0| \approx 0.27\text{--}0.29$ , the defining series converges slowly and  $K \gtrsim 15N$  is needed for reliable PSLQ convergence at the working precision. Both scripts now warn on the command line when  $K < 15N$ , so a rerun with too small a  $K$  (e.g. the  $K = 30\text{--}60$  used in some early exploratory runs) is flagged rather than silently read as a negative result.